

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOB

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'rinishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!

5.03.2026

1. Birdan farqli ikki o'zaro tub sonlarning eng kichik umumiy karralisi 225 ga teng. Bu sonlarning har biriga 3 qo'shib yangi sonlar hosil qilindi. Yangi hosil qilingan sonlarning EKUK ini toping.

A) 84
B) 48
C) 78
D) 64

2. Hisoblang:

$$(-1)^1 + (-1)^2 + (-1)^3 + \dots + (-1)^{2026} .$$

A) 1
B) -1
C) 0
D) 2

3. Ikki kombayn bilan daladagi hosilni 12 kunda yig'ib olish mumkin. Agar hosil bitta kombaynda yig'ib olinsa, u holda birinchi kombayn ikkinchisiga qaraganda 10 kun ortiq ishlagan bo'lar edi. Bu hosilni birinchi kombayn necha kunda yig'ib oladi?

A) 20
B) 30
C) 24
D) 22

4. Sotuvchi bir molning 10% ini 50% foydasi bilan. 60% ini 20% foydasi bilan va qolganini 30% foydasi bilan sotib 1560\$ foyda oldi. Bunga ko'ra molning sotib olingan narxi necha \$ ga teng?

A) 5900
B) 5600
C) 6000
D) 6400

5. Hisoblang:

$$\frac{\left(\frac{1}{9}\right)^{-3} \cdot 81^2 \cdot \left(\frac{1}{9}\right)^4 + \left(\frac{1}{6}\right)^{-4}}{225} .$$

A) 9
B) $\frac{1}{5}$
C) 45
D) 5

6. Hisoblang:

$$\sqrt[3]{4} - \frac{1}{\sqrt[3]{16} + \sqrt[3]{12} + \sqrt[3]{9}}.$$

- A) $2\sqrt[3]{4}$
- B) $\sqrt[3]{3}$
- C) 0
- D) 1

7. Soddalashtiring:

$$\sqrt{-x^2 + 6x - 9} + \sqrt[4]{x + 13}.$$

- A) $3 - x$
- B) $x - 3$
- C) 2
- D) 1

8. Arifmetik progressiyada $a_1 + a_3 + a_5 + \dots + a_{2n-1} = 96$, $a_{n-2} + a_{n+2} = 12$ bo'lsa, n ning qiymatini toping.

- A) 8
- B) 32
- C) 24
- D) 16

9. O'suvchi geometrik progressiyaning hadlari uchun $b_1 + b_2 + b_3 = 49$ va $\frac{1}{b_1} + \frac{1}{b_2} + \frac{1}{b_3} = \frac{1}{4}$ bo'lsa, birinchi hadini toping.

- A) 6
- B) 8
- C) 7
- D) 3

10. Agar $x = -3$ va $y = 2$ bo'lsa, $\left(\frac{x}{y}\right)^x + \left(\frac{-y}{x}\right)^{-x} - (xy)^y$ ni hisoblang.

- A) -6
- B) -36
- C) $\frac{8}{27}$
- D) $\frac{27}{8}$

11. Ifodani soddalashtiring:

$$x^2 y^2 \left(\frac{1}{(x+y)^2} \cdot \left(\frac{1}{x^2} + \frac{1}{y^2} \right) + \frac{2}{(x+y)^3} \left(\frac{1}{x} + \frac{1}{y} \right) \right).$$

- A) $x + y$
- B) 1

- C) xy
D) $xy(x + y)$

12. Soddalashtiring:

$$\frac{\operatorname{tg}^3 \alpha + \operatorname{ctg}^3 \alpha}{\operatorname{tg} \alpha + \operatorname{ctg} \alpha} - \frac{\operatorname{tg}^3 \alpha - \operatorname{ctg}^3 \alpha}{\operatorname{tg} \alpha - \operatorname{ctg} \alpha}.$$

- A) $\operatorname{tg}^2 \alpha$
B) $\operatorname{ctg}^2 \alpha$
C) -2
D) $\frac{2}{\cos^2 \alpha}$

13. Tenglamalar sistemasi yechimlari (x_0, y_0) bo'lsa, $x_0 + y_0$ yig'indining qiymatini toping:

$$\begin{cases} x^{7y-4} = 125 \\ x^{2y-3} = 0,2 \end{cases}.$$

- A) 4
B) 5
C) 6
D) 7

14. Tenglamaning haqiqiy ildizlari sonini toping:

$$\frac{\sqrt{\cos(\pi x) + 1} - \log_{(x+1)}^2(x^2 - x + 1)}{\sqrt{\cos(\pi x) + 1} + \sqrt{2x^2 - 5x + 2}} = 1.$$

- A) 1
B) 2
C) 3
D) Haqiqiy ildizga ega emas

15. Tenglamaning haqiqiy ildizlari yig'indisini toping:

$$\log_{x+3}(6x^2 + 28x + 30) = 5 - 2\log_{6x+10}(x^2 + 6x + 9).$$

- A) 2
B) 4
C) 1
D) 0

16. Agar a soni $x^4 - 5x^3 + 8x^2 - 5x + 1 = 0$ tenglamaning eng katta ildizi bo'lsa, u holda a sonining butun qismini toping.

- A) 4
B) 1
C) 2
D) 3

17. Tengsizlikni qanoatlantiruvchi butun sonlar nechta:

$$\frac{|5 - 4x| - 9}{4 + |x - 2|} \leq 0 .$$

- A) 2 ta
- B) 3 ta
- C) 4 ta
- D) 5 ta

18. $\sin 2x = (\cos x - \sin x)^2$ tenglamaning $(0; 180^\circ)$ oraliqdagi ildizlari yig'indisini toping.

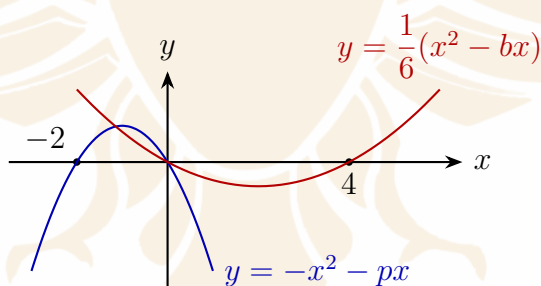
- A) 75°
- B) 105°
- C) 90°
- D) 120°

19. Tengsizlikning butun yechimlari yig'indisini toping:

$$\sqrt{6 - x^2 + 5x} < 2x + 5 .$$

- A) 22
- B) 21
- C) 20
- D) 14

20. $y = \frac{1}{6}(x^2 - bx)$ va $y = -x^2 - px$ kvadrat funksiyalar berilgan. Agar funksiyalar grafiklari $O(0; 0)$ va A nuqtalarda kesishsa A nuqta koordinatalarini toping.



- A) $\left(-\frac{10}{7}; \frac{40}{49}\right)$
- B) $\left(-\frac{9}{7}; \frac{45}{49}\right)$
- C) $\left(-\frac{6}{7}; \frac{48}{49}\right)$
- D) $\left(-\frac{8}{7}; \frac{48}{49}\right)$

21. $f(x) = \frac{x}{3} + 2$ bo'lsin. $\underbrace{f(\dots f(f(x)) \dots)}_{2026}$ funksiyaning $x = 4$ nuqtadagi qiymatini toping.

- A) $3 + 3^{-2026}$
- B) $3 + 3^{-2025}$
- C) $3 - 3^{-2026}$
- D) $3 + 3^{-2027}$

22. $f(x) = \log_3(4 + \log_3(x^2 - 3))$ bo'lsa, $f'(2)$ ning qiymatini toping.

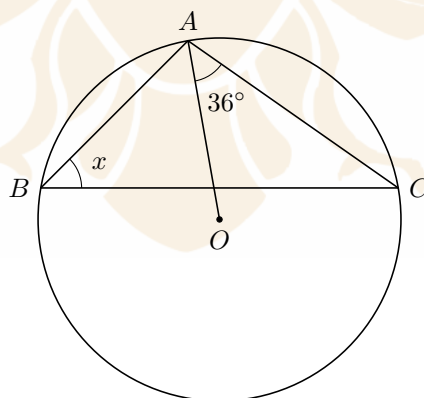
- A) $\frac{4}{\ln^2 3}$
- B) $\frac{1}{\ln^2 3}$
- C) $\frac{\ln^2 3}{4}$
- D) $\ln^2 3$

23. Aniqmas integralni toping:

$$\int \frac{x^2 + 2}{(x^4 - 3x^2 + 4) \cdot \operatorname{arctg}\left(x - \frac{2}{x}\right)} dx .$$

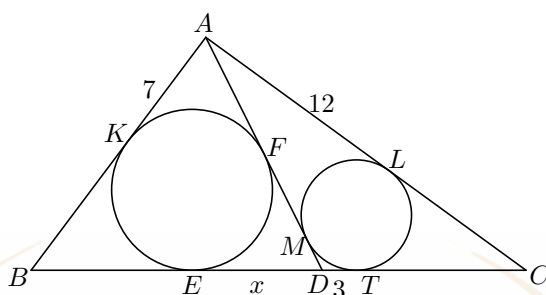
- A) $\ln \left| \operatorname{arctg}\left(x - \frac{1}{x}\right) \right| + C$
- B) $\ln \left| \operatorname{arctg}\left(x - \frac{2}{x}\right) \right| + C$
- C) $\ln \left| \operatorname{arctg}\left(x + \frac{1}{x}\right) \right| + C$
- D) $\ln \left| \operatorname{arctg}\left(x + \frac{2}{x}\right) \right| + C$

24. ABC uchburchakka O markazli aylana tashqi chizilgan. Bunda $\angle OAC = 36^\circ$ bo'lsa, $\angle ABC = x$ ning qiymatini toping.

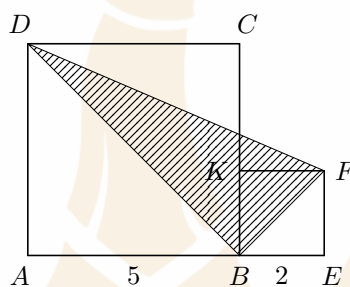


- A) 64
- B) 54
- C) 48
- D) 72

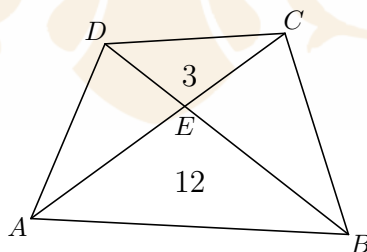
25. ABD va ADC uchburchaklarga ichki aylanalar chizilgan. $|AK| = 7$ cm, $|AL| = 12$ cm va $|DT| = 3$ cm bo'lsa, $|ED| = x$ ning qiymatini toping.



- A) 6
B) 4
C) 8
D) 5
26. $ABCD$ va $BEFK$ kvadratlar, A, B, E nuqtalar bir to'g'ri chiziqda, $|AB| = 5$ cm, $|BE| = 2$ cm. Yuqoridagi ma'lumotlarga ko'ra, BFD uchburchakning yuzini toping.

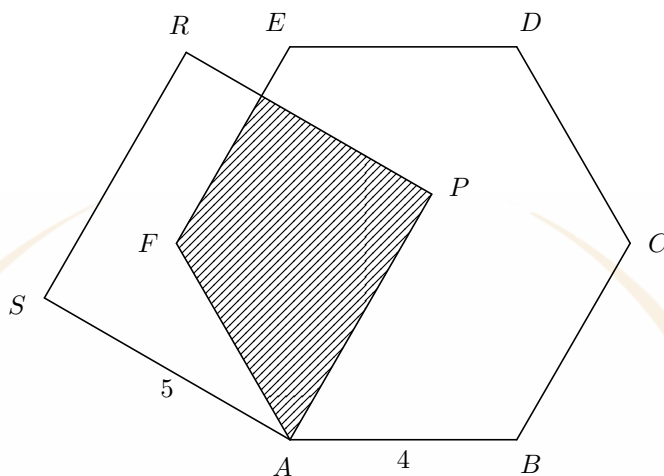


- A) 5
B) 10
C) 15
D) 20
27. $ABCD$ trapetsiya berilgan. Trapetsiya diagonallari ajratgan DEC va AEB uchburchak yuzalari mos ravishda 3 va 12 ga teng. Trapetsiya yuzini toping.

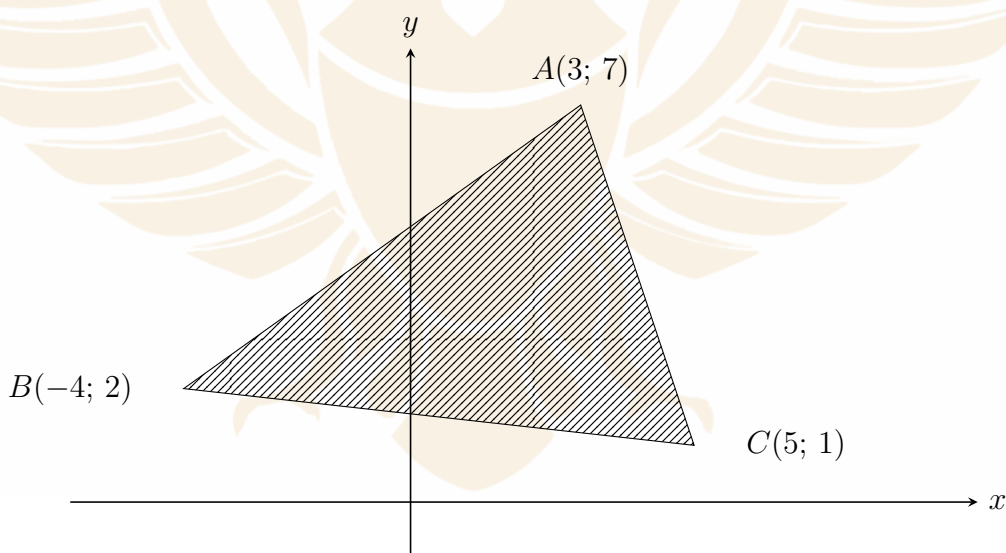


- A) 28
B) 26
C) 25
D) 27

28. $ABCDEF$ muntazam oltiburchak va $APRS$ kvadrat berilgan. Agar $AP \parallel BC$, $|AB| = 4$ cm, $|AS| = 5$ cm bo'lsa, bo'yalgan soha yuzini toping.

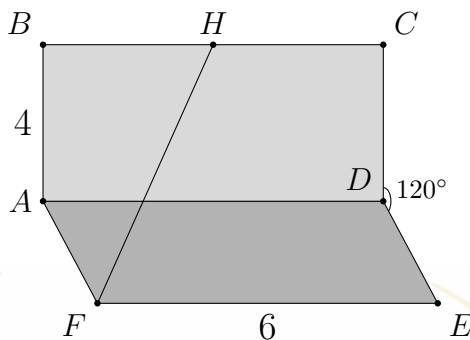


- A) $8\sqrt{3}$
 B) 12
 C) $6\sqrt{3}$
 D) 9
29. Uchlari koordinatasi $A(3; 7)$, $B(-4; 2)$ va $C(5; 1)$ nuqtalarda bo'lgan ABC uchburchakni koordinatalar boshiga nisbatan 180° ga aylantirish natijasida hosil bo'lgan uchburchakning uchlari koordinatalari yig'indisini toping.



- A) -16
 B) -15
 C) -14
 D) -13

30. Fazoda ikki tekislikka ikkita teng to'g'ri to'rtburchaklar rasmdagidek joylashtirilgan. Agar, tekisliklar orasidagi burchak 120° , $AB = 4$, $FE = 6$, $BH = HC$ bo'lsa, FH kesma uzunligini toping.



- A) $\sqrt{54}$
 B) $\sqrt{57}$
 C) 6
 D) $\sqrt{84}$
31. $A = \{a; b; c; d; e; f\}$ to'plamning nechta qism to'plamlarida b elementi bo'lib, c elementi qatnashmaydi?
 A) 16
 B) 32
 C) 8
 D) 28
32. "KOMBINATORIKA" so'zidagi harflar qog'ozga alohida yozib chiqilgan. Harflar ichidan tasodifiy bit-tasi tanlanganda, K harfi chiqish ehtimolini toping.
 A) $\frac{1}{6}$
 B) $\frac{2}{11}$
 C) $\frac{2}{13}$
 D) $\frac{2}{15}$

Topshiriqlar (33–35) va javob variant (A–F) larni o'zaro moslashtiring.

O'q kesimi kvadratdan iborat silindrga eng katta hajmli konus ichki chizilgan. Konusga esa eng katta hajmli shar ichki chizilgan. (Bunda $\pi \approx 3$ deb oling)

33. $\frac{V_{\text{konus}}}{V_{\text{silindr}}}$ ni toping.

34. $\frac{r_{\text{shar}}}{r_{\text{silindr}}}$ ni toping.

35. $\frac{V_{\text{silindr}}}{V_{\text{shar}}}$ ni toping.

Javoblar.

A) $\frac{1}{3}$

B) $\frac{\sqrt{5} + 2}{2}$

C) $\frac{\sqrt{5} - 2}{2}$

D) $\frac{2}{3}$

E) $\frac{3(\sqrt{5} + 2)}{2}$

F) $\frac{\sqrt{5} - 1}{2}$

36. Tenglamani yeching:

$$\frac{\frac{x}{1-x^2} - \frac{\sqrt{1-x^2}}{x^2}}{\frac{1}{\sqrt{1-x^2}} - \frac{1}{x}} = 9.$$

a) Tenglama nechta haqiqiy ildizga ega?

Javob: _____

b) Tenglamaning barcha ildizlari ko'paytmasini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

37. Tenglamani yeching:

$$\sin^3 x \cdot \cos 3x + \cos^3 x \cdot \sin 3x = \frac{3}{8}.$$

a) Tenglamaning eng kichik musbat ildizini toping.

Javob: _____

b) Tenglamaning $\left[-\frac{5\pi}{2}; 2\pi\right]$ oralig'idagi yechimlari sonini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

38. $f(x) = \frac{ax+4}{2x+b}$ funksiya uchun $f^{-1}(4) = 2$ va $f(x) = f^{-1}(x)$ tengliklar o'rinli.

a) $a \cdot b$ ni toping.

Javob: _____

b) Funksiyaning aniqlanish sohasiga kirmaydigan sonni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

39. $y_1 = x^2 + 2$ va $y_2 = x^2 + 4x + 5$ kvadrat funksiyalar berilgan. Funksiyalarga umumiy o'tkazilgan urinma tenglamasi $y = kx + b$ bo'lsin.

a) $k \cdot b$ ning qiymatini toping.

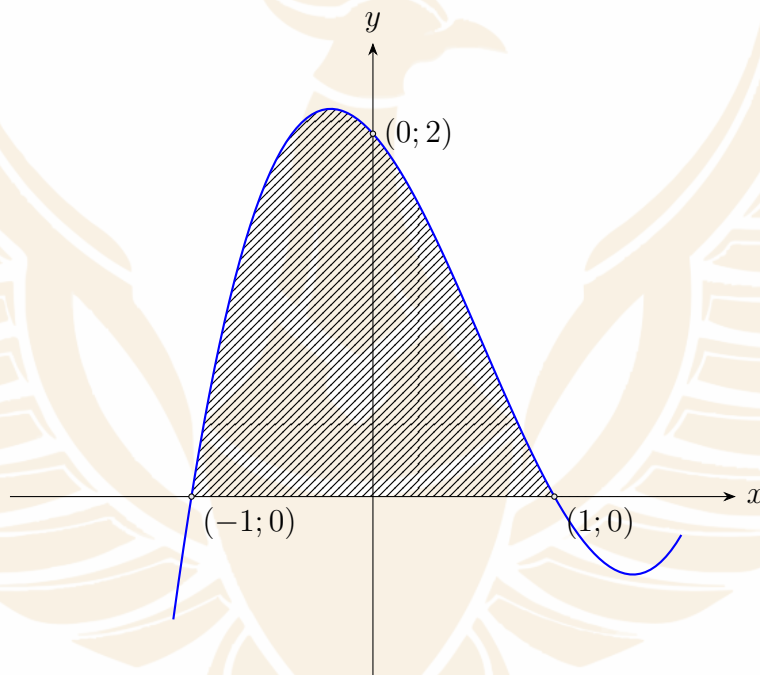
Javob: _____

b) Urinish nuqtalari orasidagi masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

40. $f(x) = ax^3 + bx^2 + cx + d$ funksiya $(-1; 0)$, $(0; 2)$ va $(1; 0)$ nuqtalardan o'tadi.



a) $\frac{a + b + c}{d}$ ning qiymatini toping.

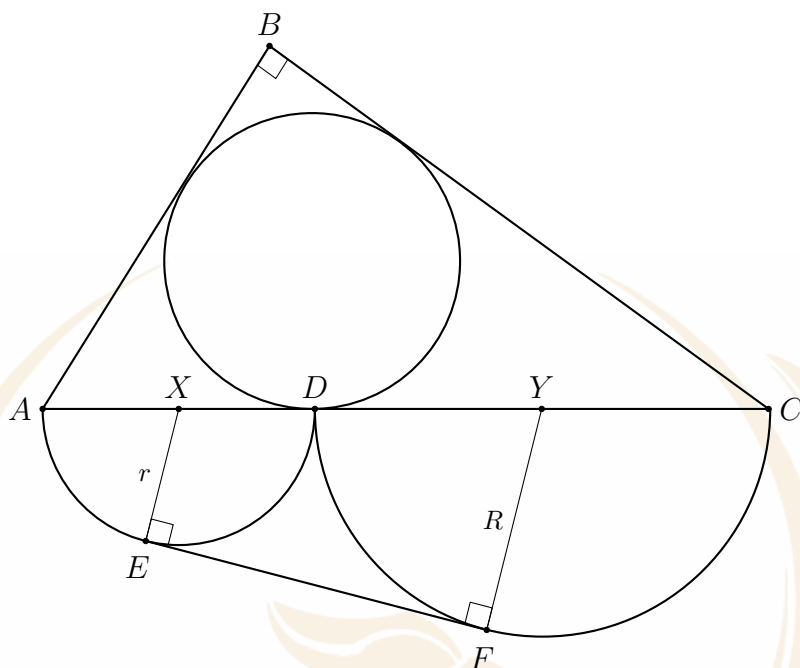
Javob: _____

b) Bo'yalgan soha yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

41. ABC to'g'ri burchakli uchburchakka ichki aylana va AC gipotenuzasini diametr qilib yarim radiuslari r, R bo'lgan yarim aylanalar chizilgan. $EF = 3\sqrt{3}$ bo'lsin.



a) $r \cdot R$ ni toping.

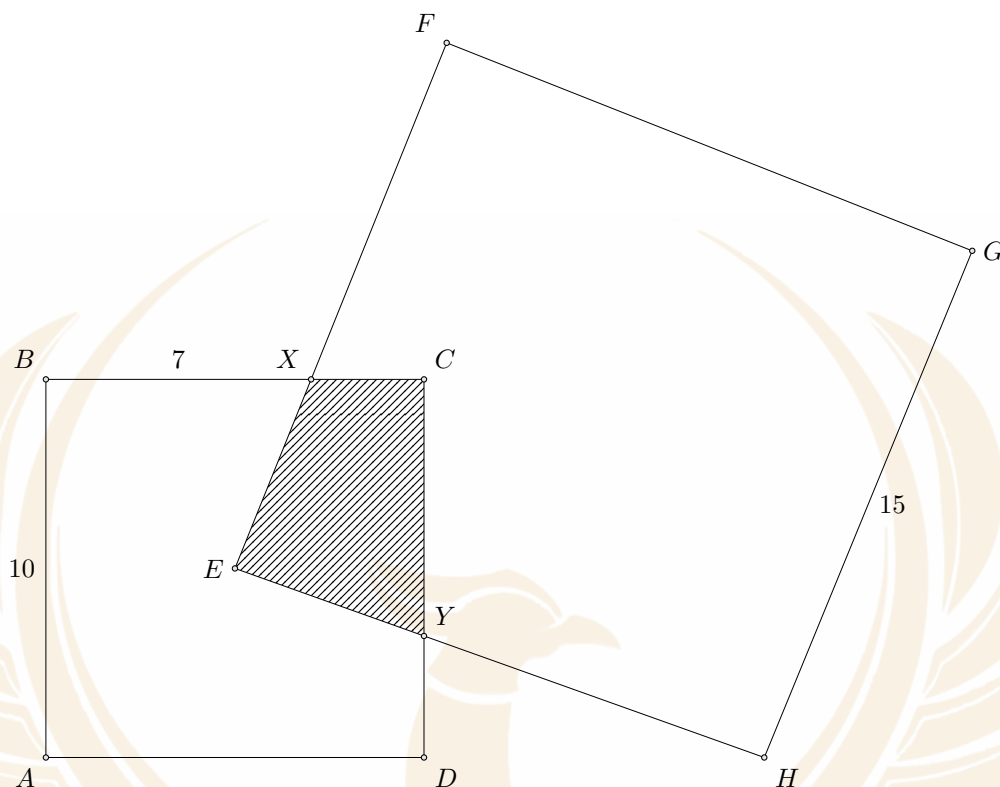
Javob: _____

b) S_{ABC} ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

42. $ABCD$ tomoni 10 ga teng kvadrat bo'lsin. Mayli $EFGH$ tomoni 15 ga teng kvadrat bo'lsin. E nuqta $ABCD$ kvadratning markazi. EF tomon BC tomonni X nuqtada kesadi va EH tomon CD ni Y nuqtada kesadi. Agar $BX = 7$ bo'lsa,



- a) $EXCY$ to'rtburchakning yuzasini toping.

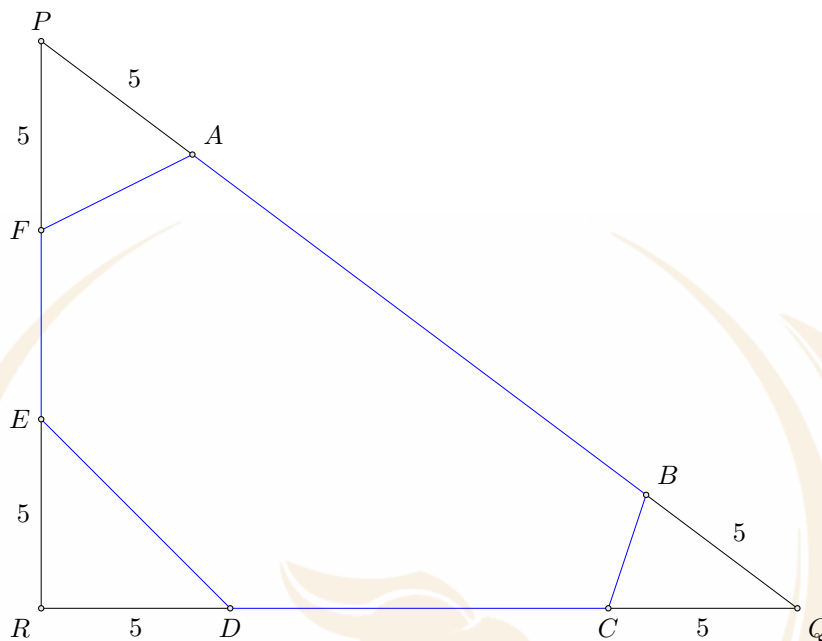
Javob: _____

- b) $ABXEYD$ oltiburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

43. $\triangle PQR$ to'g'ri burchakli uchburchakda $PR = 15$, $QR = 20$ va $PQ = 25$. A va B nuqtalar $\overline{PQ}$ da, C va D nuqtalar $\overline{QR}$ da, E va F nuqtalar $\overline{PR}$ tomonlarda yotadi. Bunda $PA = QB = QC = RD = RE = PF = 5$ tenglik o'rinli bo'lsa,



- a) PFA va QBC uchburchaklar yuzalarining musbat farqini toping.

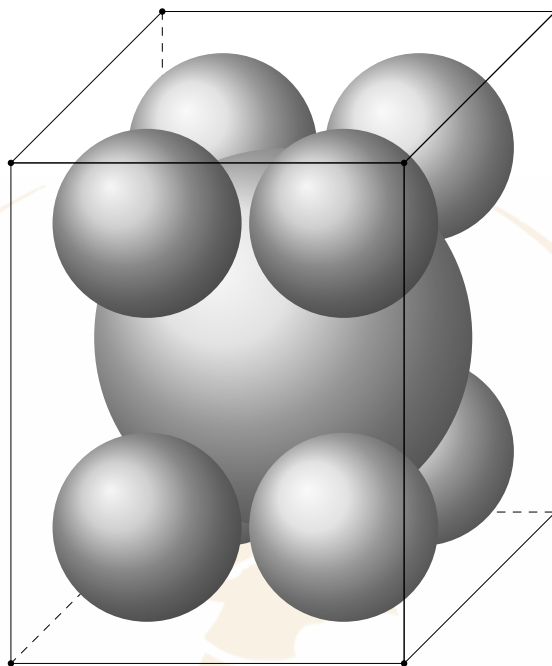
Javob: _____

- b) $ABCDEF$ oltiburchakning yuzasini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ga ko'chirib yozing.

44. O'lchamlari $4 \times 4 \times h$ bo'lgan to'g'ri burchakli usti ochiq quti ichida radiusi 2 bo'lgan bitta shar va radiusi 1 bo'lgan sakkizta kichik shar joylashgan (rasmga qarang). Kichik sharlarning har biri qutining uch tomoniga urinadi va katta shar har bir kichik sharga urinadi. Agar h ni $a + b\sqrt{c}$ ko'rinishida ifodalash mumkin bo'lsa, (Bunda $\sqrt{7} \approx 2,5$, $\pi \approx 3$ deb oling.)



a) $a + b + c$ ni toping.

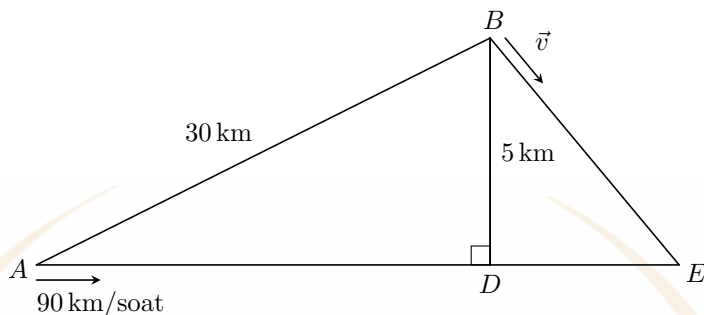
Javob: _____

b) Qutiga shu holatda suv solinmoqda, quti to'lishi uchun qutiga qancha suv quyish kerak.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

45. Quyida A nuqtadan bir avtobus gorizantal yoʻnalishda 90 km/h tezlik bilan yoʻlga chiqmoqda. Xuddi shu vaqtda B nuqtadan bir otliq avtobusni quvib yetish uchun yoʻlga chiqdi. $AB = 30 \text{ km}$, $BD = 5 \text{ km}$ boʻlsin. Otliq avtobusga E nuqtaga yetib bormoqchi.



a) Otliqning tezligi eng kamida necha km/h boʻlishi kerak?

Javob: _____

b) AE masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga koʻchirib yozing.